A PSEUDO-COHERENT COMPLEX OVER A FINITE FREE ALGEBRA HAS A BOUNDED ABOVE COMPLEX OF FINITE FREE MODULES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The cohomology of the complex is a finite module over the base ring in the degrees where it is nonzero, because the complex is pseudo-coherent over that ring. invoked at proof lines 8694-8698. Produces the finite module whose surjection from a finite free module starts the construction.
- In a distinguished triangle, if two terms are pseudo-coherent then so is the third, which lets the cone of the comparison map be used in the induction. invoked at proof lines 8720-8728. Keeps the cone pseudo-coherent over the base ring so that the induction hypothesis applies to it.

Source reference. Original lemma 064S: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 8694-8698 for the finiteness of the lowest nonvanishing cohomology; the choice of the finite free module built from it is inside the counted span.

Source reference. Original definition 014E: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 8720-8722 to form the cone; the induction step carried out with that cone is inside the counted span.

Source reference. Original lemma 064R: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 8723-8728 to see that the cone is again pseudo-coherent over the base ring; the induction hypothesis applied to it is inside the counted span.

Lemma 0.2. *Let A be a ring. Let R be a (possibly noncommutative) A-algebra which is finite free as an A-module. Then any object M of $D(R)$ which is pseudo-coherent in $D(A)$ can be represented by a bounded above complex of finite free (right) R-modules.*

Proof. Choose a complex $M^\bullet$ of right R-modules representing M. Since M is pseudo-coherent we have $H^i(M) = 0$ for large enough i. Let m be the smallest index such that $H^m(M)$ is nonzero. Then $H^m(M)$ is a finite A-module by More on Algebra, Lemma 064S. Thus we can choose a finite free R-module F^m and a map

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$F^m \to M^m$ such that $F^m \to M^m \to M^{m+1}$ is zero and such that $F^m \to H^m(M)$ is surjective. Picture:

$$\begin{array}{ccccccc} & & F^m & \longrightarrow & 0 & \longrightarrow & \ldots \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow & & \\ M^{m-1} & \longrightarrow & M^m & \longrightarrow & M^{m+1} & \longrightarrow & \ldots \end{array}$$

By descending induction on $n \leq m$ we are going to construct finite free R-modules F^i for $i \geq n$, differentials $d^i : F^i \to F^{i+1}$ for $i \geq n$, maps $\alpha : F^i \to K^i$ compatible with differentials, such that (1) $H^i(\alpha)$ is an isomorphism for $i > n$ and surjective for $i = n$, and (2) $F^i = 0$ for $i > m$. Picture

$$\begin{array}{ccccccccccc} & & F^n & \longrightarrow & F^{n+1} & \longrightarrow & \ldots & \longrightarrow & F^i & \longrightarrow & 0 & \longrightarrow & \ldots \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha} & & & & \downarrow{\scriptstyle \alpha} & & \downarrow & & \\ M^{n-1} & \longrightarrow & M^n & \longrightarrow & M^{n+1} & \longrightarrow & \ldots & \longrightarrow & M^i & \longrightarrow & M^{i+1} & \longrightarrow & \ldots \end{array}$$

The base case is $n = m$ which we've done above. Induction step. Let $C^\bullet$ be the cone on α (Derived Categories, Definition 014E). The long exact sequence of cohomology shows that $H^i(C^\bullet) = 0$ for $i \geq n$. Observe that $F^\bullet$ is pseudo-coherent as a complex of A-modules because R is finite free as an A-module. Hence by More on Algebra, Lemma 064R we see that $C^\bullet$ is $(n-1)$-pseudo-coherent as a complex of A-modules. By More on Algebra, Lemma 064S we see that $H^{n-1}(C^\bullet)$ is a finite A-module. Choose a finite free R-module F^{n-1} and a map $\beta : F^{n-1} \to C^{n-1}$ such that the composition $F^{n-1} \to C^{n-1} \to C^n$ is zero and such that F^{n-1} surjects onto $H^{n-1}(C^\bullet)$. Since $C^{n-1} = M^{n-1} \oplus F^n$ we can write $\beta = (\alpha^{n-1}, -d^{n-1})$. The vanishing of the composition $F^{n-1} \to C^{n-1} \to C^n$ implies these maps fit into a morphism of complexes

$$\begin{array}{ccccccc} & & F^{n-1} & \xrightarrow[d^{n-1}]{} & F^n & \longrightarrow & F^{n+1} & \longrightarrow & \ldots \\ & & \downarrow{\scriptstyle \alpha^{n-1}} & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha} & & \\ \ldots & \longrightarrow & M^{n-1} & \longrightarrow & M^n & \longrightarrow & M^{n+1} & \longrightarrow & \ldots \end{array}$$

Moreover, these maps define a morphism of distinguished triangles

$$\begin{array}{ccccccc} (F^n \to \ldots) & \longrightarrow & (F^{n-1} \to \ldots) & \longrightarrow & F^{n-1} & \longrightarrow & (F^n \to \ldots)[1] \\ \downarrow & & \downarrow & & \downarrow{\scriptstyle \beta} & & \downarrow \\ (F^n \to \ldots) & \longrightarrow & M^\bullet & \longrightarrow & C^\bullet & \longrightarrow & (F^n \to \ldots)[1] \end{array}$$

Hence our choice of β implies that the map of complexes $(F^{n-1} \to \ldots) \to M^\bullet$ induces an isomorphism on cohomology in degrees $\geq n$ and a surjection in degree $n-1$. This finishes the proof of the lemma. $\square$